Error Correction Method for Smart Energy Meter Field Calibration System under Non-standard Conditions

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***Abstract*—Smart energy meters have been widely used in the smart grid of various countries, but the current measurement methods of smart energy meters can not solve the traceability problem under the actual load current and power factor in the process of use, which leads to the inaccuracy of only meters in the field environment, and there is an unknown measurement risk. In order to achieve the accurate on-site verification of smart energy meters in normal working conditions, this paper studies the accuracy of smart energy meter field calibration standard device under different current and power factors, develops the error correction method under non-standard conditions. The correction method adopts the method of establishing reference standard value arrays under non-standard conditions to achieve fast and accurate on-site verification and effective measurement performance evaluation of the smart energy meters in use, to explore the measurement work for on-site calibration of smart energy meters under current and power factor in the actual environment.**

Keywords-component; smart energy meters; field calibrator; measurement; error correction; data surface

I. INTRODUCTION

China consumes 7 trillion kilowatt-hours of electricity every year. By improving the smart energy meter field calibration device, the production quality of the smart energy meter and the overall energy measurement level of the power grid can be improved. Every 1% increase in the overall energy metering level will save 7 billion kilowatt-hours of electricity, which is equivalent to half a year of the Three Gorges Hydropower Station station's annual power generation. The improvement of the on-site measurement level of smart energy meters can provide technical guarantee for the trade settlement of electric energy, improve public satisfaction, promote the improvement of the quality of meter products, and strongly support the construction of smart grids.

Worldwide, the quality and accuracy of smart grids are directly related to energy conservation and emissions reduction and sustainable human development. Smart power meters are the core and foundation of smart grid measurement and settlement and energy monitoring. At present, the measurement and verification of smart energy meter by OIML is generally carried out in the ideal environment of the laboratory [1]. For the smart energy meter in use, the on-site verification is generally to bring the standard meter to the site, connect the standard meter to the grid close to the meter to be tested, and under the same or similar load conditions, compare the electric energy readings of the meter to be tested and the standard meter in the same time, so as to get the error of the meter to be tested. When the standard meter is connected to the power grid, the current loop of the standard meter will be connected in series in the load circuit, and various interferences of the power grid will have a great impact on the measurement. Moreover, it is impossible to compare and measure various load conditions [2].

In recent years, with the increasing intensity of on-site monitoring and verification of the energy meter, in order to accurately detect the energy meter in the work without affecting the user's use, the use of the energy meter on-site calibration standard device for on-site continuous electrical detection has become the needs of the new era. The smart power meter field verification device is a measuring device used to verify the smart energy meter on site. Its accuracy and reliability are directly related to the fairness and fairness of electric energy trade settlement. The accuracy of the standard device directly determines the on-site verification result of the smart energy meter. The high-precision standard device can more accurately detect out-of-tolerance smart energy meters and replace it to improve the accuracy of the overall energy measurement. The accuracy of energy measurement is related to the user's measurement and settlement and national energy monitoring. The smart power meter field verification device provides early warning and early rotation of the metering performance failure of the smart energy meter, which extends the service life of the meter with good metering performance and scientific flexibility to avoid waste and protect the interests of people.

According to the current metrological verification regulations, smart power meters have many different standard measurement points, but there are a lot of non-standard conditions during the field test time. At present, it is difficult to evaluate the positive and negative of the overall smart energy meter for the measurement data under non-standard conditions deviation. It is necessary to establish a reference standard value array under non-standard conditions, so as to improve the accuracy of on-site verification. Develop the corresponding error correction method for smart power meter field verification device under non-standard

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conditions to realize the automation of on-site calibration with high accuracy.

Based on the long-term monitoring of the smart energy meters verification data, the developed temperature controlled high-precision remote smart power meter field verification device. According to the uncertainty characteristics of current and power factor in field detection, the standard data surface of two-dimensional nonlinear influence quantity is built by interpolation, and then the smooth interpolation surface is obtained by Multi-Quadric(MQ) method. The standard data surface of two-dimensional nonlinear influence quantity of load, power factor and electric energy error is built, so that the measurement error can be corrected under any current and power factor. Use the smart power meter in the field to verify the operation, and correct the detection data, so as to effectively evaluate the energy measurement performance online [3].

II. Establishment of Reference Standard Value Array under Nonstandard Condition

The smart energy meter field calibrator is mainly used in the use site of smart energy meter. The actual current, power factor and ambient temperature vary greatly compared with the laboratory. In the field detection environment, these conditions are not constant, and the current will change with the user's use of equipment, and the temperature will vary greatly with the seasons.

There may be large machines in the field of the energy meter, and the current gap with the small electrical equipment is large. The power factor and current of different equipment is also different. Its own heat will further increase the working environment temperature of the energy meter. Therefore, the design of the energy meter takes into account the harshness of the use environment, and there are strict requirements for the measurement of the energy meter under different currents, power factors, and temperatures [4].

As a testing device, the field calibrator also needs to correct the error under different currents and power factors, otherwise the accuracy of its measurement results will be greatly affected [5]. After integrated precision temperature control, the working temperature of the on-site calibration device core part can be controlled to 45 ℃, which reduces the measurement error caused by the environmental temperature on site. Therefore, in the temperature-controlled field calibration standard device for smart power meter, the reference standard value array to correct the error under the non-standard condition can greatly improve the measurement accuracy.

In the laboratory, the temperature-controlled field calibration standard device for smart power meter is tested. According to the actual use environment, 9 main different power factors and 9 load points are selected, and 81 discrete errors are obtained after multiple tests, then, through interpolation, a two-dimensional nonlinear influence quantity standard data surface is established.

$$R_n(x) = f(x) - L_n(x) = \frac{f^{(n+1)}(\xi_n)}{(n+1)!}\omega_{n+1}(x)$$

$$x_k = \cos\left(\frac{(2k+1)\pi}{2(n+1)}\right) \tag{1}$$

First, use Chebyshev (formula 1) nodes [6] to process the data. At nodes with a high rate of change, denser nodes will be obtained, and at the same time, edge errors can be reduced, as shown in Fig. 1.

It is common to use evenly distributed points as the base point of the interpolation polynomial. In many cases, the data points used for interpolation only exist in this form, for example, when the data is composed of data read together at the same time interval. In other cases, we can freely choose the base point where we think fit. Facts have proved that the method of selecting the base point spacing has a great influence on the interpolation error, and Chebyshev interpolation is a specific optimal point spacing selection method. The motivation of Chebyshev interpolation is to improve the control of the maximum interpolation error in the interpolation interval. In general, Chebyshev interpolation is a specific optimal point spacing selection method. Compared with the Newton difference quotient formula, it can avoid the Runge phenomenon more effectively, and it can better fit the original Function, that is, its result prediction will be relatively more accurate.

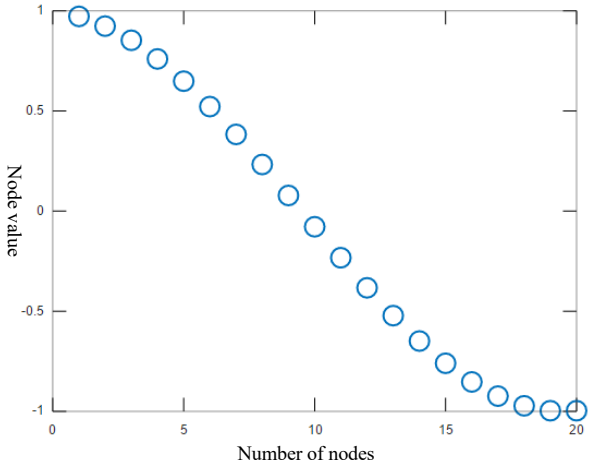


Figure 1. Chebyshev node processing data

Then, the data is processed by the gravity Lagrange interpolation polynomial (formula 2)[7]. This method does not have high requirements on the boundary and distribution of the original data. It is suitable for processing scattered data to approximate irregular continuous surfaces, which is beneficial to numerical realization and approximation effects. It is very suitable for processing the error data of the field calibration standard device for smart power meter. After processing, a total of 400 discrete points of 20×20 are obtained.

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Gravity Lagrangian interpolation does not need to calculate polynomial when it is substituted into x value calculation. The advantage of it is that when the Chebyshev node is used for interpolation, the given function can be simulated well, so that when the number of interpolation points tends to infinity, the maximum deviation tends to zero. At the same time, the gravity Lagrangian interpolation combined with the Chebyshev node for interpolation can achieve excellent numerical stability. The first type Lagrangian interpolation is backward stable, while the second type Lagrangian interpolation is forward stable, and the Lebesgue constant is small.

$$\ell_j(x) = \frac{\ell(x)}{x - x_j} \frac{1}{\prod_{i=0, i\neq j}^{k}(x_j - x_i)}$$

$$w_j = \frac{1}{\prod_{i=0, i\neq j}^{k}(x_j - x_i)} \tag{2}$$

$$L(x) = \ell(x) \sum_{j=0}^{k} \frac{w_j}{x - x_j} y_j$$

Finally, the Multi-Quadric (MQ) method is used to obtain a smooth interpolation surface (Equation 3)[8]. The MQ interpolation method does not need to solve linear equations, it does not have the stability problem, and the calculation speed is faster. field calibration standard device for smart power meter.

$$f(\boldsymbol{x}) = \sum a_j g(\|\boldsymbol{x} - \boldsymbol{x}_j\|_2) \tag{3}$$

$$g(\|\boldsymbol{x} - \boldsymbol{x}_j\|_2) = (x - x_j)^2 + (y - y_j)^2 + (z - z_j)^2 + r_j^2$$

Thus, a standard data surface of two-dimensional nonlinear influence of load, power factor and energy error is established.

In the experiment, 9 main different power factors and 9 load points were selected, and 81 discrete errors were obtained as shown in Fig. 2.

Then, after the current interpolation, it reaches 9×20 error points, as shown in Fig. 3.

After the power factor interpolation is obtained, 20×20 points are obtained, for a total of 400 error points. As shown in Fig. 4.

Finally, the Multi-Quadric method is used to process the data to obtain a smooth two-dimensional nonlinear influence standard data surface. As shown in Fig. 5.

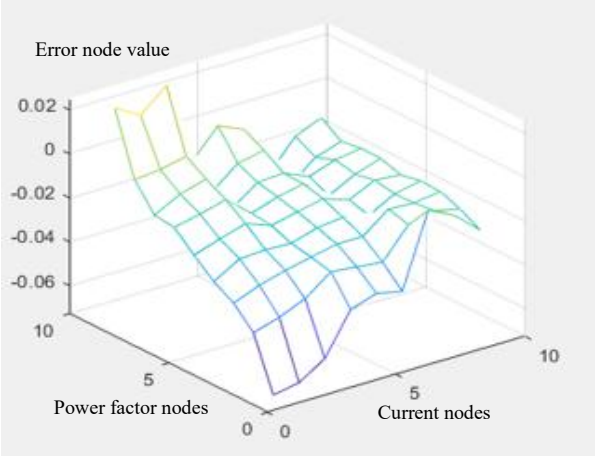


Figure 2. 81 original nodes

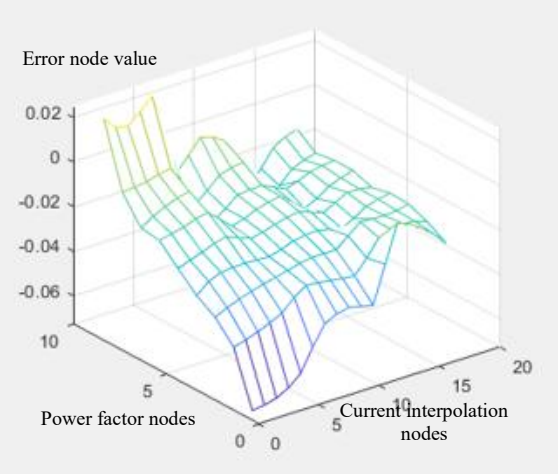


Figure 3. 9×20 error points after current interpolation

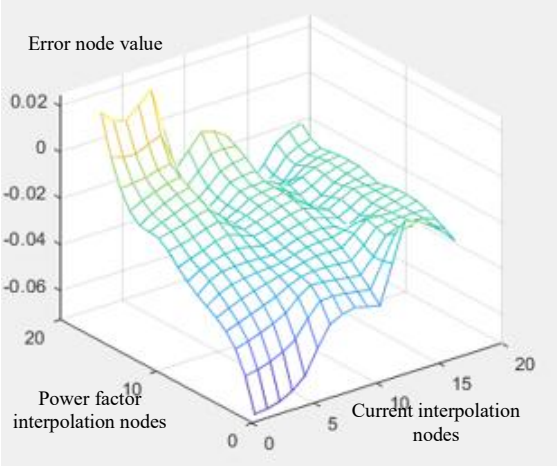


Figure 4. 20×20 error points after power factor interpolation

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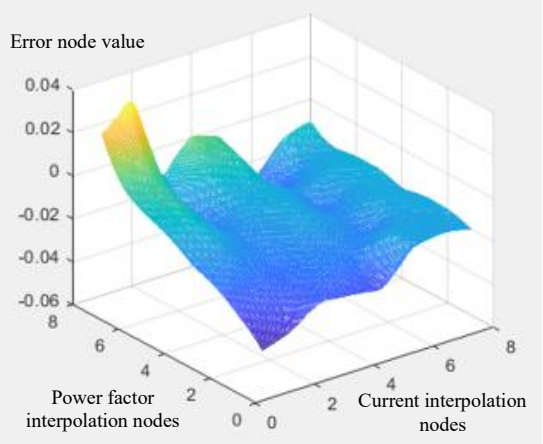


Figure 5. Smooth difference surface

In the final smooth error surface, under any load and power factor, the reference standard value can be found in the surface for traceability. At the same time, the temperature-controlled smart energy meter field calibration device can maintain a constant temperature, so that the value traceability of the ambient temperature and the complex field conditions of different field currents and power factors can be completed at the same time.The introduction of the error correction function of the reference standard value array under non-standard conditions can reduce the overall error of the smart energy meter field calibration device. The current correction method of the smart energy meter field calibration device is to correct a limited number of points, while this method incorporates countless points in the entire calibration range into the correction range. In theory, any load point within the correction range has an accurate correction value. This is a revolutionary improvement for the development of smart energy meter field calibration device.

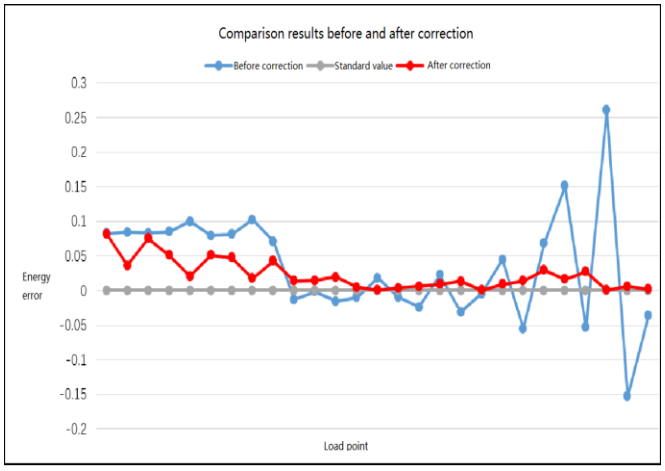


Figure 6. Field calibrator comparison results

III. VERIFICATION

Compared with the traditional field calibrator, the field calibration device with reference standard value array error correction function under non-standard conditions has a greater accuracy advantage. Under different load conditions, the measurement uncertainty after correction can reach 0.013% (k=2), and the measurement performance index is better than U=0.015% (k=2) before correction. (As shown in Fig. 7, red: The energy error of smart energy meter field calibration device after correction; blue: The energy error of smart energy meter field calibration device before correction; gray: standard value)

IV. CONCLUSION

The standard data surface of the established nonlinear influence quantity under any load and power factor, we can find the reference standard value in the surface for traceability, which lays the foundation for realizing high-level calibration in the field. The temperature-controlled smart energy meter field calibration device based on the correction of the reference standard value array error under non-standard conditions has strengthened the pertinence and effectiveness of the smart energy meter field verification, monitor the life cycle measurement performance of smart meters, improve the smart grid macro-level energy measurement level and reduce the overall measurement error, promote smart grid energy monitoring and measurement, increased efficiency and saved energy, effectively guarantee the authenticity and reliability of the measurement data. This method has a good promotion significance, it has the role of exploration and reference for the supervision of other energy measuring instruments, and will have a demonstration effect for the measurement supervision work.

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